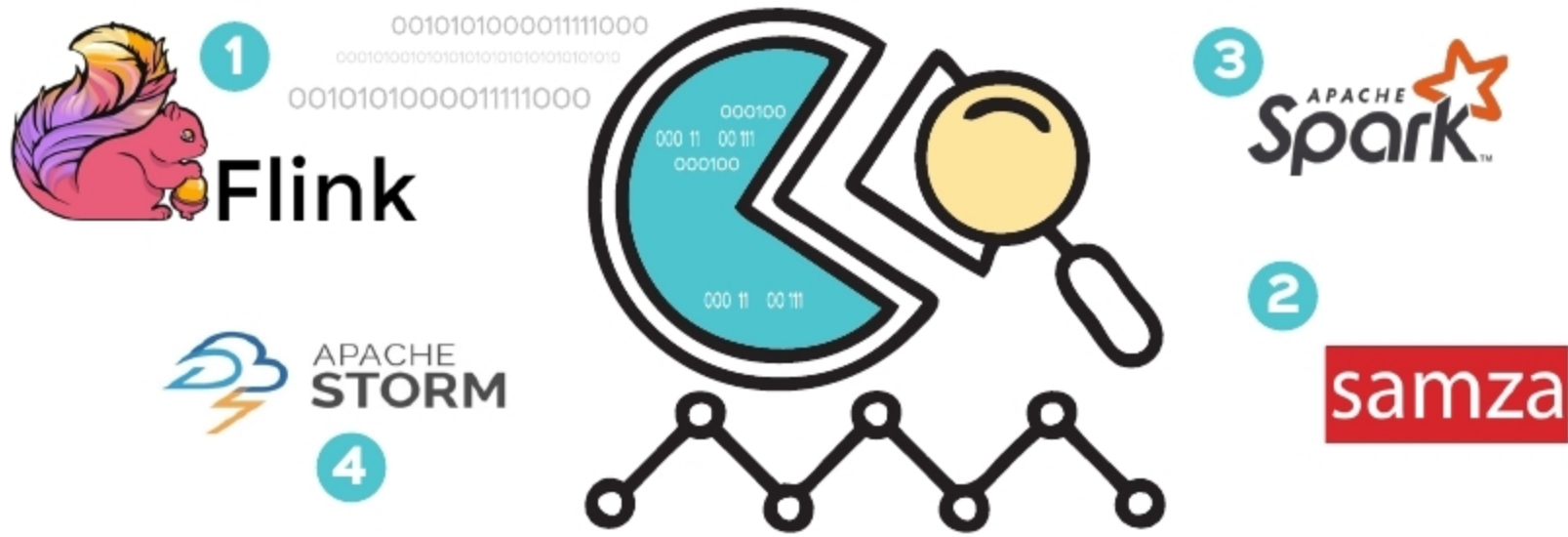


Open Source Tools that Enable Real-Time IoT Data Analytics

Abundant new data is generated every day as billions of devices are connected, thanks to the Internet of Things. This data has useful, novel, hidden insights which are very helpful for any business organisation to capitalise on. Analytics on this data helps to detect anomalies, mitigate any disruption in services, predict problems early and provide better satisfaction to customers.



Data analytics is vital for the success of IoT projects and businesses. IoT applications have several characteristics that are suitable for analytics.

1. **Volume:** IoT projects generate a large amount of data every day. Analysis of this data set reveals interesting patterns that help organisations make wise and timely decisions.
2. **Variety of data:** IoT applications capture diverse types of data such as structured, semi-structured and unstructured. Analytics is the only way to derive useful and novel insights from such a variety of data.
3. **Real-time decision-making:** Many IoT applications generate real-time data. Analytics of such real-time data is essential for fast and accurate decision making. Analytics serves as a lifesaving solution for many IoT projects.
4. **Business growth:** Businesses use the data generated by IoT for revenue generation. Analytics gives

a number of insights into the large volume of data that can really help businesses foresee trends, helping them to strategise and increase their earnings.

Types of analytics for IoT

The traditional way of analytics uses a combination of reporting and predictive analytics, but IoT applications have different characteristics so this way is not effective. IoT analytics requires real-time streaming data analysis. It demands more data, and needs automation and integration to perform efficiently.

- **Streaming analytics:** According to Jerry Baulier, vice president, Internet of Things R&D, SAS, “Streaming analytics generates informed decisions in milliseconds from millions of devices, and also examines thousands of events per second.” Streaming analytics helps in providing security by identifying events that are vulnerable to threats

and risks rapidly. It plays a key role in data driven organisations, and allows them to build real-time solutions using IoT. Real-time streaming analytics uses Big Data to generate statistics and visualisations to measure the efficacy of any process in the organisation.

Here are several open source tools for Streaming analytics.

- I. **Apache Flink:** This open source platform is capable of distributed data processing and batch data processing. Any kind of data is produced as a stream of events. Credit card transactions, sensor measurements, machine logs, user interactions on a website, or mobile applications — all of this data is generated as a stream, and can be processed as unbounded or bounded streams. Apache Flink excels at processing unbounded and bounded data sets.

